MIL-W-36775A(DLA-DM)
28 February 1979
SUPERSEDING
MIL-W-36775 (DSA-DM)
27 August 1971

MILITARY SPECIFICATION

WET BULB-GLOBE TEMPERATURE KIT

This limited coordination military specification has been prepared by the Defense Personnel Support Center based upon currently available technical information. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1 SCOPE

1.1 Scope. This specification covers a small lightweight hand-held wet bulb-globe temperature kit for providing an index of heat stress in outdoor areas of hot and humid climates, in order to control and prevent heat casualties during troop training exercises The heat stress index is computed by the weighted addition of readings obtained from three different thermometers.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

P-C-436	- Cleaning Compound, Alkali, Boiling Vat.
TT-E-527	- Enamel, Alkyd, Lustreless.
TT-L-50	- Lacquer, Nitrocellulose, Acrylic and Acrylic-Butyrate, Aerosol (in Pressurized Dispensers).
TT-P-1757	- Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN. Directorate of Medical Materiel, CODE-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearaing at the end of this document or by letter.

FSC 6665

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PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Fiberboard.
PPP-B-676 - Boxes, Setup.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-171 - Finishing of Metal and Wood Surfaces.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

25072 - Wet Bulb Globe Temperature Kit

(Copies of specifications, standards and drawing required by contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation E77-66 - Verification and Calibration of Liquid-in-Glass Thermometers.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

DEPARTMENT OF THE ARMY TECHNICAL BULLETIN

TB-MED-175 - The Etiology, Prevention, Diagnosis, and Treatment of Adverse Effect of Heat.

(For information purposes only)

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3. REQUIREMENTS

3.1 Materials and components. The materials and components shall be suitable for the intended purposes and shall conform to the requirements of DPSC Drawing No. 25072 and the associated Bill of Materials.

3.1.1 Thermometer graduations. All thermometers, within their specified ranges, shall be graduated in 1 degree F. subdivisions and numbered at least at every 10°F markings. When tested as specified, the accuracy of the thermometers used for black-globe, dry bulb, and wet bulb temperature indication shall be ± 1°F. (see 4.4.1).

3.1.2 Slide rule calculations. When tested as specified, the WBGT slide rule readings shall agree with table IV (see 4.4.2).

3.2 Design and construction The kit shall consist of a thermometer assembly and a slide rule assembly that is contained in a portable case. The kit shall be in strict accordance with DPSC Drawing for style, dimensions and construction. In addition, identification marking per 3.6 shall be included on each kit. The kit shall weigh no more than 20 ounces.

3.2.1 Finish and color.

3.2 1.1 Hinged box and cover. The hinged aluminum box and cover shall be treated according to P-C-436 prior to painting The outside of the box and cover including the hinge shall be primed according to TT-P-1757 and shall be finished with two coats of baked lusterless enamel according to TT-E-527. The color shall be olive drab approximating Federal Standard 595 Color No. 34087.

3.2 1.2 Sunshade. The sunshade shall be treated and painted as specified in 3.2.1 1. The color shall be white The lettering and markings on the sunshade shall be black.

3.2.1.3 Sheath, radiant energy. The radiant energy sheath shall be colored black using a flat lacquer which conforms to TT-L-50.

3.3 Performance. The wet bulb-globe temperature kit shall operate in accordance with the instructions included in 3.5.

3 4 Accessories. The following additional replacement parts shall be included with each kit.

1 each transparent perforated shield.
1 each water reservoir.
18 inch extra wick.

The replacement parts shall be in accordance with DPSC Drawing No. 25072 and Bill of Materials and shall be contained in the case above the cushion mat.

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3.5 Service data. Each kit shall be furnished with two copies of service data which specifically includes the following instructions and figure (i.e., see figure 1).

GENERAL INSTRUCTIONS

WET BULB-GLOBE TEMPERATURE KIT

NSN 6665-00-159-2218

A The wet bulb-globe temperature kit is an instrument for providing information on hot weather risks to the health of troops undergoing training. The information is displayed on a scale in the form of an index, computed from the weighed readings obtained from three different thermometers.

1. A stationary wet bulb (WB) thermometer exposed to the sun and prevailing wind.

2. A similarly exposed "black globe" (BG) thermometer with a black sheath over the bulb. The sheath and bulb are inside a transparent perforated plastic shield.

3. A dry bulb (DB) thermometer with its bulb shielded from the direct rays of the sun by a shield painted white.

B. The wet bulb-globe temperature index (WBGT index) is equal to 70% of the wet bulb (WB) temperature reading plus 20% of the black globe (BG) temperature reading plus 10% of the dry bulb (DB) temperature reading and can be computed as follows:

$$\text{WBGT Index} = 0.7\ \text{WB} + 0.2\ \text{BG} + 0.1\ \text{DB}$$

Use of the attached slide rule provides the correct WBGT index automatically.

C. To use the WBGT index as a control of physical activity, the following guide lines from TB-MED-175 are provided.

1. When the WBGT index reaches 82, discretion should be used in planning exercise for unseasoned personnel.

2. When the WBGT index reaches 85, strenuous exercises such as marching at standard cadence should be suspended for unseasoned personnel during their first three weeks of training. At this WBGT index, training activities may be continued on a reduced scale after the second week of training.

3. Outdoor classes in the sun should be avoided when the WBGT index exceeds 85.

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4. When the WBGT index reaches 88, strenuous exercise should be curtailed for all recruits and other trainees with less than 12 weeks training in hot weather. Hardened personnel after having been acclimatized each season, can carry on limited activity at WBGT indices of 88 to 90 for periods not exceeding six hours a day.

D. The WBGT kit is enclosed in an aluminum case. The case is kept closed with a miniature stainless steel catch. The threaded hole in the bottom of the case is used to attach the case to a standard lightweight Photographers tripod that is not supplied with this kit.

1. The kit is opened by disengaging the catch and lifting the cover.

2. Lift the thermometer assembly up and out (see figure 1).

NOTE Examine the column of each thermometer. If the liquid has separated, heat the bulb slowly and carefully until the liquid reunites. CAUTION. Never use a match or any kind of open flame to heat a thermometer bulb. Use a warm liquid only.

3 Wet the wet bulb wick thoroughly. NOTE The wick is cotton and is 5 inches long with a knot 1 inch from one end The end nearer the knot is pushed into the reservoir and the other end is slipped on the thermometer bulb (see figure 1). The water reservoir should be filled with clear, preferable deionized or distilled water and utilized as indicated The water should be changed daily and the wick washed with soap and rinsed thoroughly. To avoid error in the measurement of the wet bulb temperature, the water in the reservoir must be free of salt and soap.

4. Hold the kit with the thermometers toward the sun, with the "black globe" thermometer closest to the sun. Wait 10 minutes for stabilization of temperatures.

5. Review the instructions on the right side of the slide rule (WET BULB-GLOBE TEMPERATURE INDEX CALCULATOR). Assume for purposes of instruction that the Black Globe (BG) temperature reading is 120, the Wet Bulb (WB) temperature reading is 80, and the Dry Bulb (DB) temperature is 100.

(a) Move 80 on the Wet Bulb temperature (WB TEMP) scale so that it is directly under 100 on the Dry Bulb temperature scale (DB TEMP).

(b) Find 120 on the Black Globe temperature scale (BG TEMP).

(c) Read Wet Bulb Globe Temperature Index (WBGT INDEX) at the Black Globe temperature (BG TEMP).

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(d) If you have performed the slide rule movement correctly, the index should read 90.

E CAUTION. After use, empty the water reservoir to prevent wetting or rusting of parts in the kit.

F The kit contains in addition to the parts shown in figure 1, the following spare components:

1 each transparent perforated plastic shield.
1 each water reservoir.

3.6 Identification marking. The following identification marking shall be included on the outside of the case.

Manufacturer's name or registered trademark.
Contract or purchase order number.
National Stock Number (NSN).
Item Identification.
U.S.

The identification marking shall be either stamped, printed, or engraved on the case or shall be in the form of waterproof decal with permanent type ink, which is firmly and securely attached to the case. All marking shall be legible.

3.7 Workmanship. Workmanship shall be first class throughout. The kit shall be free from defects which detract from its appearance or impair its serviceability.

3.7.1 Welding. All welds shall be durable and sound in every detail, with all joints cleaned, and exposed joints ground and finished smooth to blend with adjacent areas.

3.7.2 Thermometers. The graduation markings shall be complete, permanent and legible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements including the technical requirements for the manufacturer's part numbers specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

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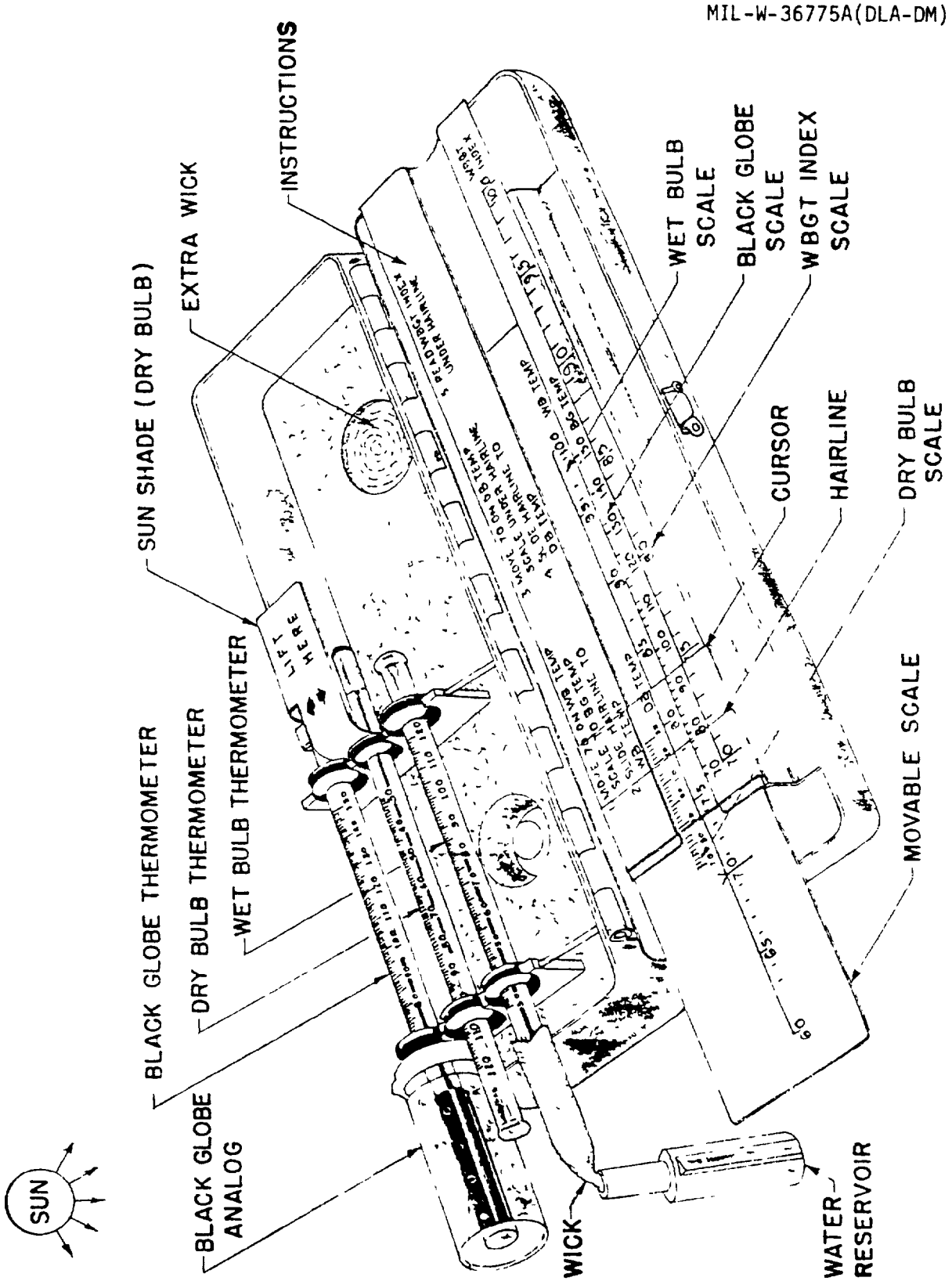


FIGURE 1. Wet Bulb-Globe Temperature Kit

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4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection Inspection, as used in this specification, is defined as both examination (such as visual, or auditory, investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the thermometers, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificates shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing. the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for sampling purposes shall be the kit.

TABLE I. Sampling for examination.

	Inspection level	AQL (Percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-1	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II. The unit of product for sampling purposes shall be as indicated in the table.

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TABLE II. Sampling for tests.

	Characteristic	Requirement	Test procedure	Inspection level	AQL
Thermometer	Verification of: Accuracy of calibration	3.1.1	4 4.1	S-2	1 0
Slide rule	Accuracy of calibration	3.1.2	4.4.2	S-2	*

* Acceptance shall be zero regardless of lot size.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects shown in table III. Examination shall not be restricted to the possible defects that are listed in the table.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Kit not complete with all component parts.
102	Components not free of damage, dents or deformation which renders them unsuitable for intended use.
103	Thermometers not capable of being secured within holders.
104	Thermometer assembly not capable of being cammed or maintained in the raised position.
105	Thermometer assembly not capable of being properly seated within the case.
106	Any weld not sound or not free of being broken or cracked.
107	Slide rule not free of warping or deformation, or not easily adjustable throughout the entire range.
108	Marking on slide rule not complete, permanent or legible.
109	Case not capable of being properly opened, closed, or securely latched.
110	Finish on cylinder around "black globe" thermometer not complete or not free of chipping.
111	Graduation markings on thermometers not complete, permanent, or legible.
Minor	
201	Identification marking not complete, permanent, or legible.
202	Finish on case or slide rule not free of chipped areas, scratches or roughness.
203	Exposed surfaces of case not free of burrs or sharpness.

4.3.2 Dimensional examination. The kit shall be examined for defects in dimensions. Any dimension not within the tolerance specified shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.3.2 Dimensional examination. The kit shall be examined for defects in dimensions. Any dimension not within the tolerance specified shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Verification of accuracy of thermometer calibration. Samples of each type thermometer shall be tested in accordance with ASTM Designation E77-66 for calibration. The thermometers shall be tested at three temperature points, which are the end and middle points.

4.4.2 Accuracy of slide rule calculation. The slide rule shall be used to determine the WBGT based on data in table IV. The calculated WBGT shall be compared with the value indicated in table IV.

TABLE IV.

BG	WB	DB	WBGT
90	79	80	75
100	80	90	85
120	90	100	97

4.4.3 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Packaging Packaging shall be level A or C, as specified (see 6.1)

5.1.1 Unit of issue Each wet bulb-globe temperature kit constitutes a unit of issue.

5.1.2 Level A.

5 1.2.1 Unit packaging. Each unit along with accessories and replacement parts shall be clean, dry and positioned within the case. The case shall be latched and packaged in a box conforming to PPP-B-566 or PPP-B-676 Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C Each unit along with accessories and replacement parts shall be clean, dry and positioned within the case. The case shall be further packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the first receiving activity.

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5.2 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.2.1 Level A. Ten unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Ten unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged kits shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the carriers' rules and regulations.

5.4 Marking. Unit packages and shipping containers shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following

(a) Title, number and date of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

6.3 This specification covers the following item listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6665-00-159-2218	WET BULB-GLOBE TEMPERATURE KIT

Preparing activity:
Defense Logistics Agency - DM

Project Number 6665-0390

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE
MIL-W-36775A(DLA-DM) WET BULB-GLOBE TEMPERATURE KIT

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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